

HW 3: Custom ROS Wheeled Robot

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Abstract—This assignment focused on modifying a URDF robot using xacro files and controlling the robot using a C++ ROS program. The robot was modified into a wheeled design by replacing the cylindrical arms with wheels. A ROS publisher node was implemented to control robot movement in Gazebo simulation.

Keywords—ROS, URDF, Xacro, Gazebo, C++, mobile robotics, robot modeling, robotic control, simulation, ROS publisher

I. INTRODUCTION

The objective of this assignment was to modify an existing URDF robot model and implement a C++ ROS program to control its movement in Gazebo.

II. METHODOLOGY

A. URDF Modification

The original robot consisted of a cube body, cylindrical arms, and a camera head. The cylindrical arms were modified into wheels by adjusting their geometry and joint configuration in the xacro files. The original robot model shown in Fig. 1 was taken from the Training (ii) – URDF lecture video [1].

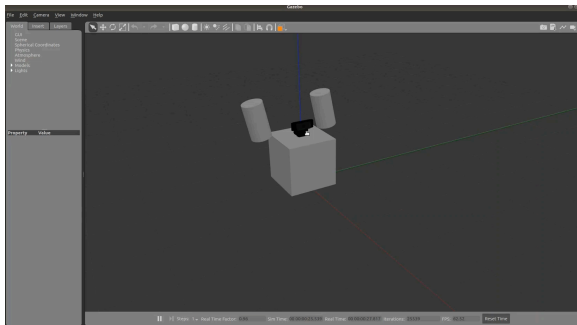


Fig. 1. Original Robot Model (source: [1]).

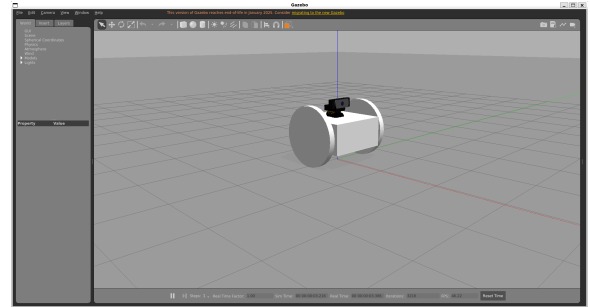


Fig. 2. Modified Wheeled Robot in Gazebo Isometric View.

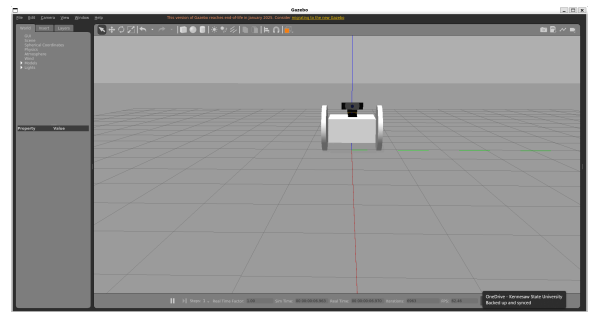


Fig. 3. Modified Wheeled Robot in Gazebo Side View.

B. C++ Control Program

A ROS C++ publisher node was created to send std_msgs/Float64 commands to the robot's arm controllers. This allowed control of wheel rotation using ROS topics.

```
1 #include <iostream>
2 #include <ros/ros.h>
3 #include <std_msgs/Float64.h>
4
5 int main(int argc, char** argv)
6 {
7     ros::init(argc, argv, "my_robot_node");
8     ros::NodeHandle n;
9
10    ros::Publisher leftPub = n.advertise<std_msgs::Float64>("robot_arm_controller/command", 1000);
11    ros::Publisher rightPub = n.advertise<std_msgs::Float64>("robot_arm_controller/command", 1000);
12    ros::Rate loop_rate(10);
13
14    while (ros::ok()) {
15        std::cout << "Please enter a value:" << std::endl;
16        std::cin >> val;
17        std_msgs::Float64 speedMsg;
18        speedMsg.data = -1.0;
19        leftPub.publish(speedMsg);
20        rightPub.publish(speedMsg);
21    }
22    return 0;
23 }
```

Fig. 4. C++ Publisher Code (main.cpp).

C. Robot Motion

The robot was tested in Gazebo and successfully responded to ROS commands, demonstrating forward and backward movement.

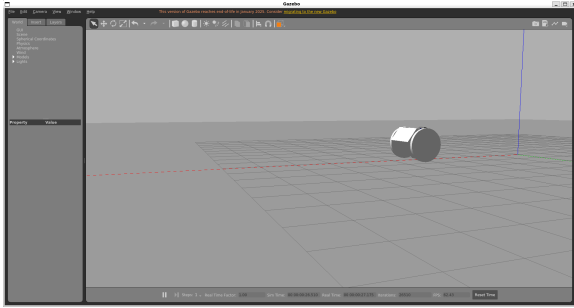


Fig. 5. Robot Movement in Gazebo.

III. RESULTS

The robot was successfully modified into a wheeled robot and controlled using a ROS C++ publisher node. The robot responded correctly to movement commands in Gazebo.

IV. CONCLUSION

The assignment was completed successfully by modifying the URDF model and implementing a ROS C++ program to control robot motion in simulation.

V. REFERENCES

- [1] M. H. Tanveer, "Training (ii) – URDF," course lecture video, Department of Robotics & Mechatronics Engineering, Kennesaw State University, accessed Jul. 2026.
- [2] UTRA RoboSoccer, "Tutorials-2020 ROS Robot Simulation Repository," GitHub. <https://github.com/utra-robosoccer/Tutorials-2020>. Accessed: Jul. 2026.
- [3] ROS Wiki, "Writing a Simple Publisher and Subscriber (C++)," ROS Tutorials. <https://wiki.ros.org/ROS/Tutorials/WritingPublisherSubscriber%28c%2B%2B%29>. Accessed: Jul. 2026.